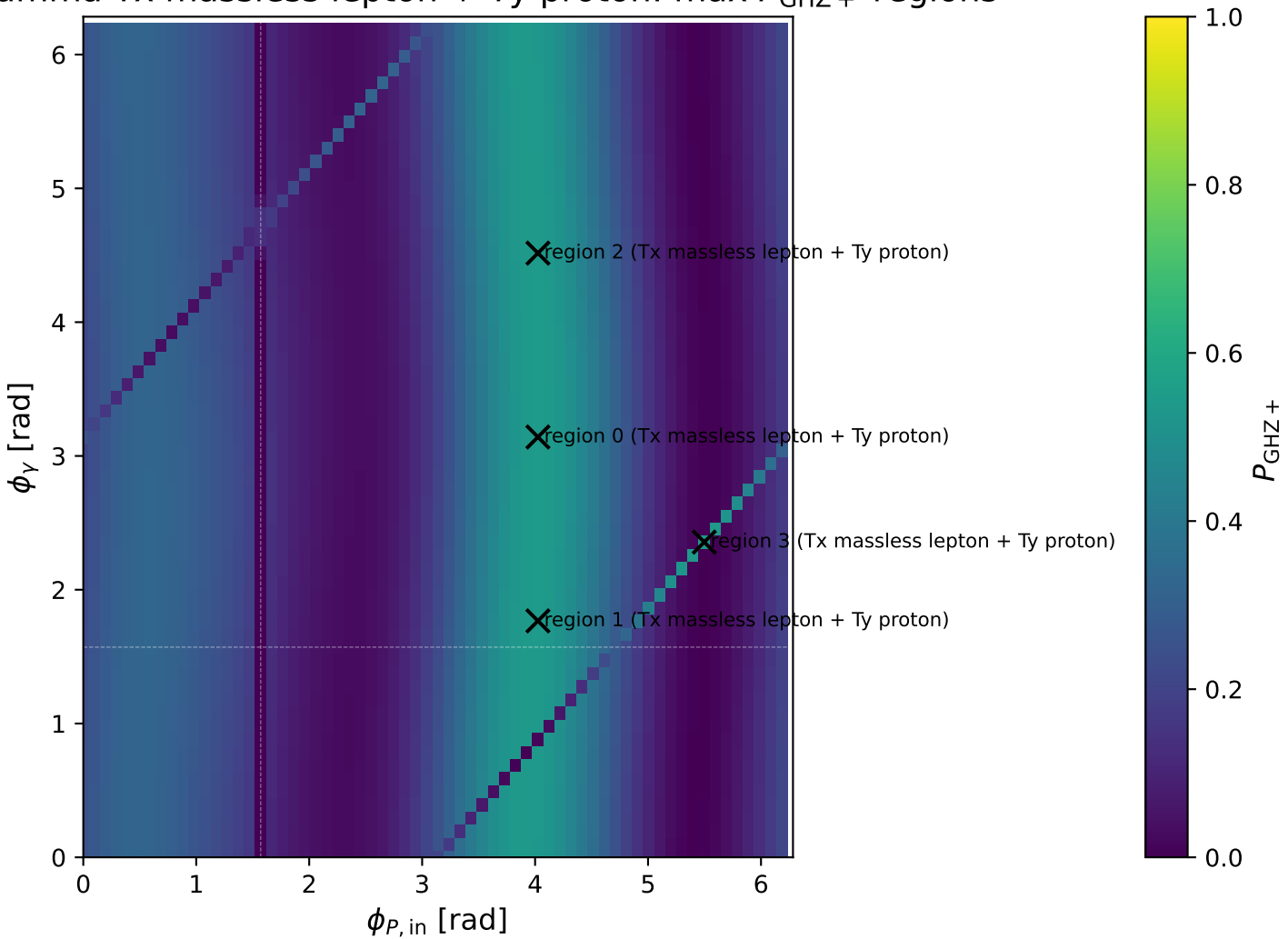
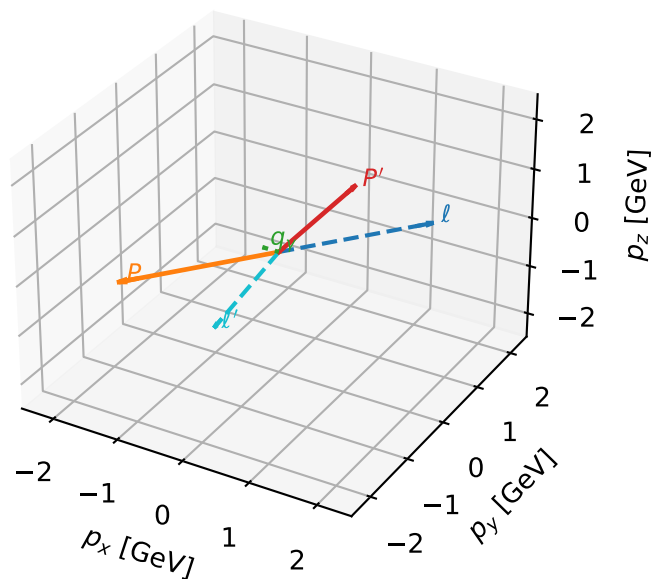


gamma Tx massless lepton + Ty proton: max $P_{\text{GHZ}} +$ regions



Tx massless lepton + Ty proton characteristic max P_{GHZ} + configuration

3D momenta

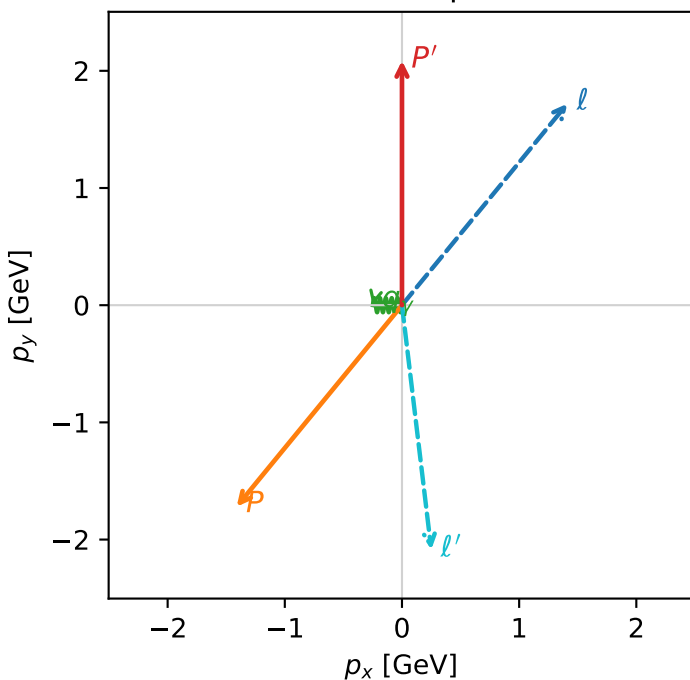


ghz_purity_low_egamma_txtty_region_0 $P_{\text{GHZ}}=0.545632$
 region: low_Egamma_TxTy_region_0
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_low_Egamma
 incoming state: $|T_{x,\ell_0}\rangle \otimes |\bar{T}_{y,p}\rangle$

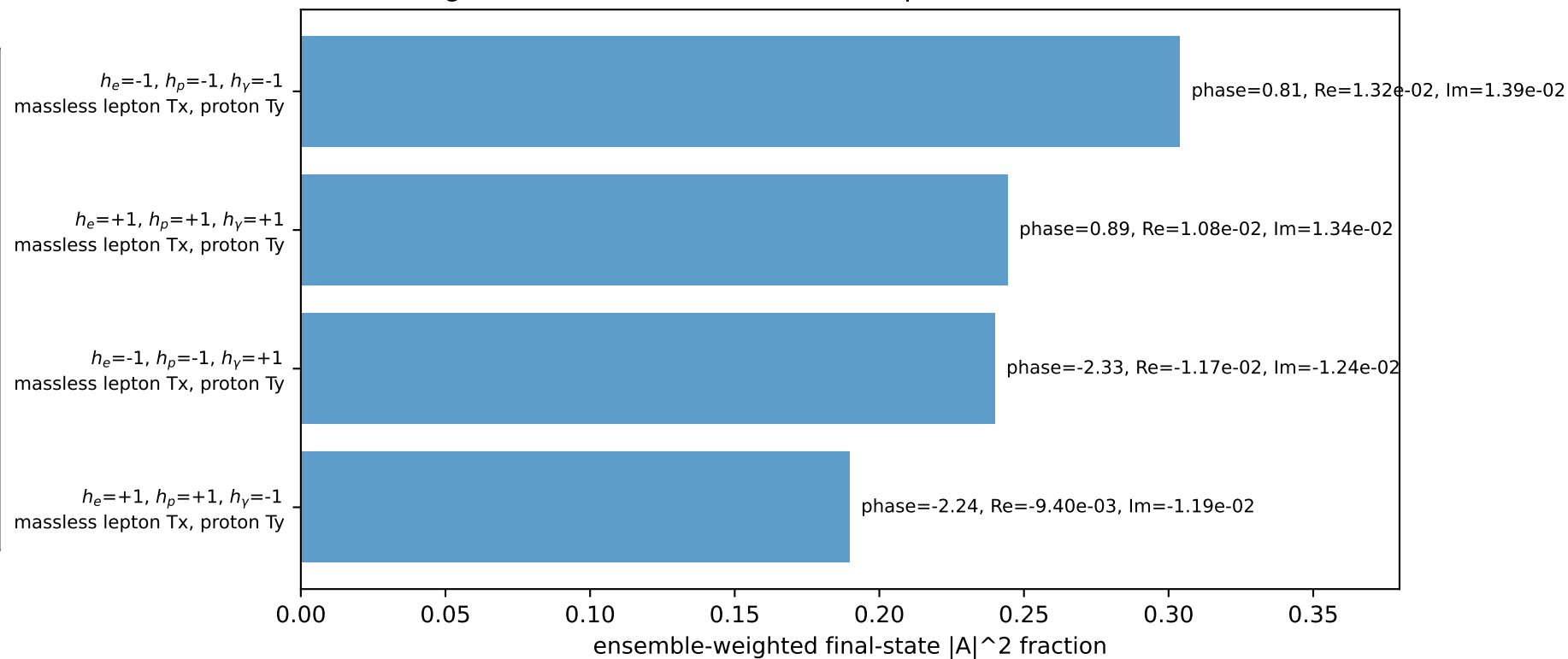
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=2.08621$
 $\theta_{\text{in}}=1.57079$, $\phi_{\ell}=0.883573$, $\phi_p=4.02517$
 $E_{\gamma}=0.25$, $\phi_{\gamma}=3.14159$
 $Q^2=15.8165$, $x_B=0.932566$, $t=-16.4587$, $W^2=2.02353$, $y=0.821874$
 $\theta(\ell', \gamma)=96.8334$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2244$ $p_x= 1.4112$ $p_y= 1.7195$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= -1.4112$ $p_y= -1.7195$ $p_z= 0.0000$
 ℓ' $E= 2.1011$ $p_x= 0.2500$ $p_y= -2.0862$ $p_z= 0.0000$
 P' $E= 2.2874$ $p_x= 0.0000$ $p_y= 2.0862$ $p_z= 0.0000$
 q_{γ} $E= 0.2500$ $p_x= -0.2500$ $p_y= 0.0000$ $p_z= 0.0000$

Transverse plane

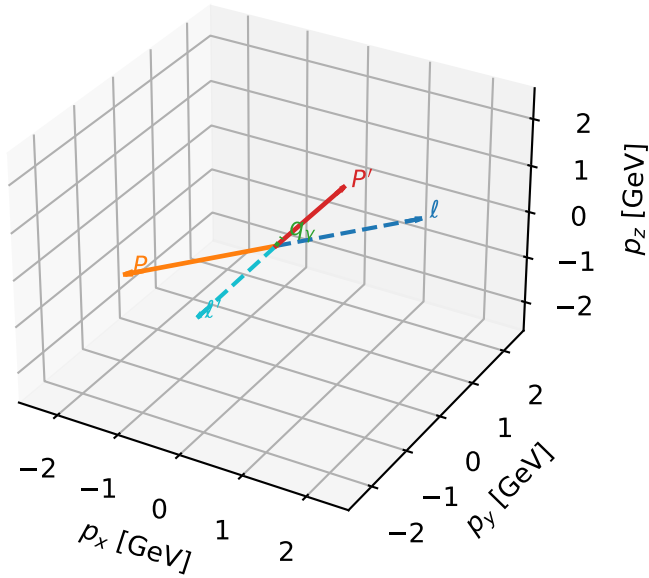


Leading initial-ensemble/final-state components (N=4, retained=97.8%)



Tx massless lepton + Ty proton characteristic max P_{GHZ} + configuration

3D momenta

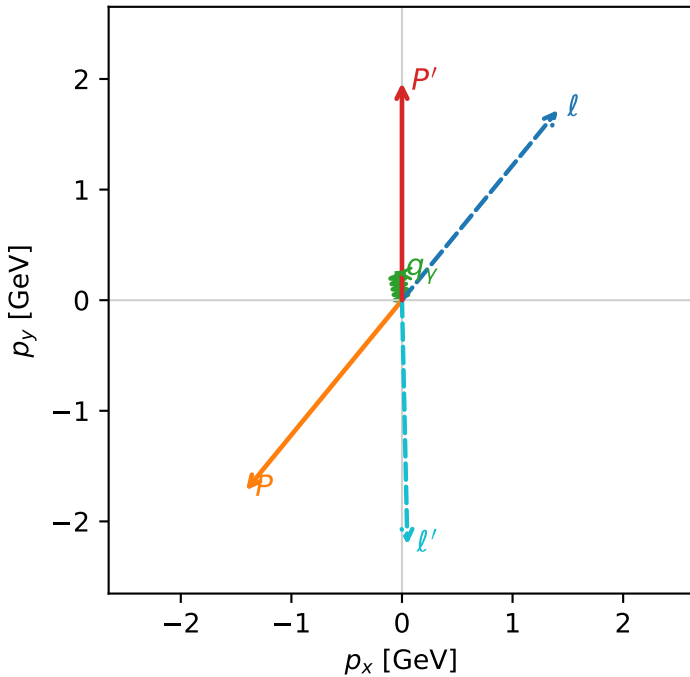


ghz_purity_low_egamma_txtty_region_1 $P_{\text{GHZ}}=0.543748$
 region: low_Egamma_TxTy_region_1
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_low_Egamma
 incoming state: $|T_{x,\ell_0}\rangle \otimes |\bar{T}_{y,p}\rangle$

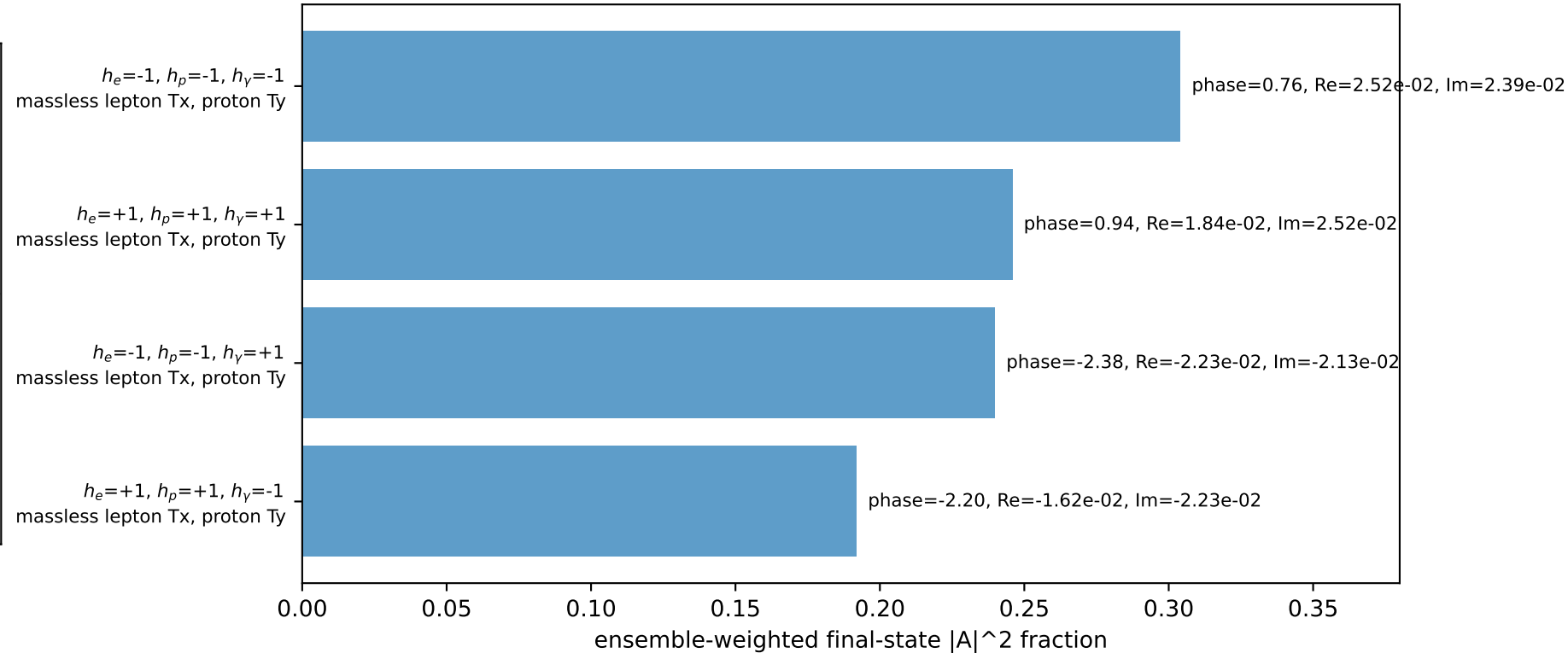
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=1.9652$
 $\theta_{\text{in}}=1.57079$, $\phi_{\ell}=0.883573$, $\phi_p=4.02517$
 $E_{\gamma}=0.25$, $\phi_{\gamma}=1.76715$
 $Q^2=17.3$, $x_B=0.992822$, $t=-15.5124$, $W^2=1.00491$, $y=0.844401$
 $\theta(\ell', \gamma)=170.014$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2244$ $p_x= 1.4112$ $p_y= 1.7195$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= -1.4112$ $p_y= -1.7195$ $p_z= 0.0000$
 ℓ' $E= 2.2109$ $p_x= 0.0488$ $p_y= -2.2104$ $p_z= 0.0000$
 P' $E= 2.1776$ $p_x= 0.0000$ $p_y= 1.9652$ $p_z= 0.0000$
 q_{γ} $E= 0.2500$ $p_x= -0.0488$ $p_y= 0.2452$ $p_z= 0.0000$

Transverse plane

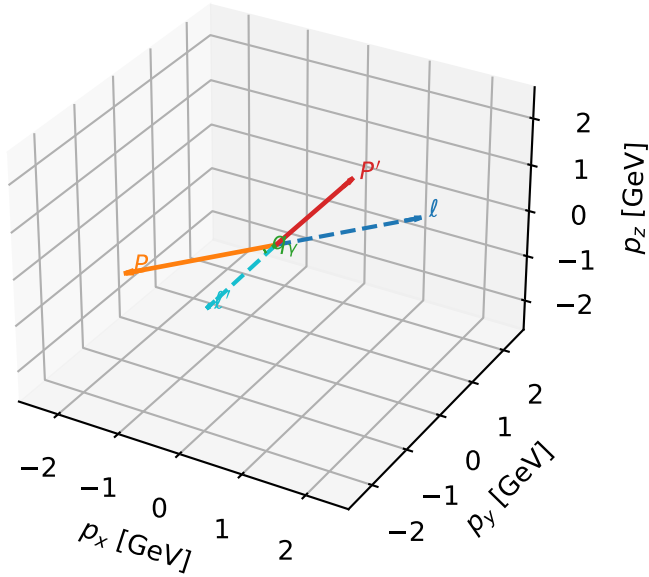


Leading initial-ensemble/final-state components (N=4, retained=98.1%)



Tx massless lepton + Ty proton characteristic max P_{GHZ} + configuration

3D momenta

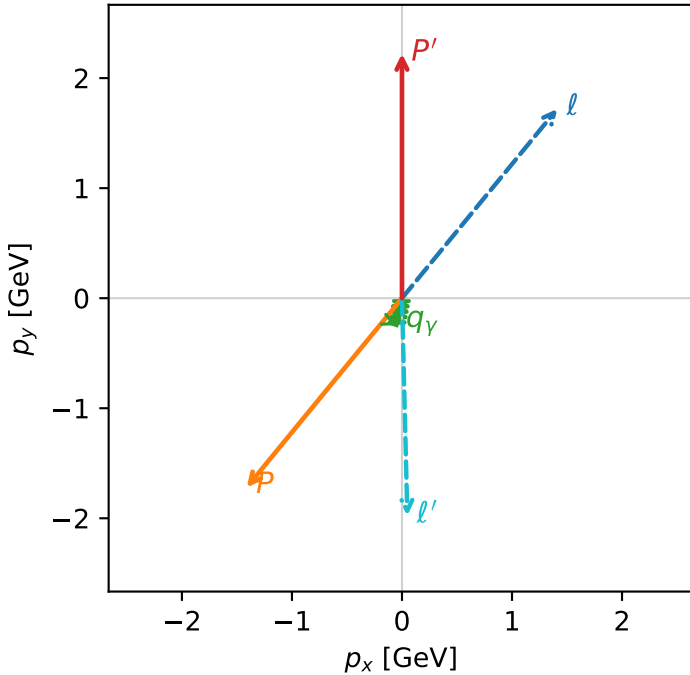


ghz_purity_low_egamma_txtty_region_2 $P_{\text{GHZ}}=0.543435$
 region: low_Egamma_TxTy_region_2
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_low_Egamma
 incoming state: $|T_{x,\ell_0}\rangle \otimes |\bar{T}_{y,p}\rangle$

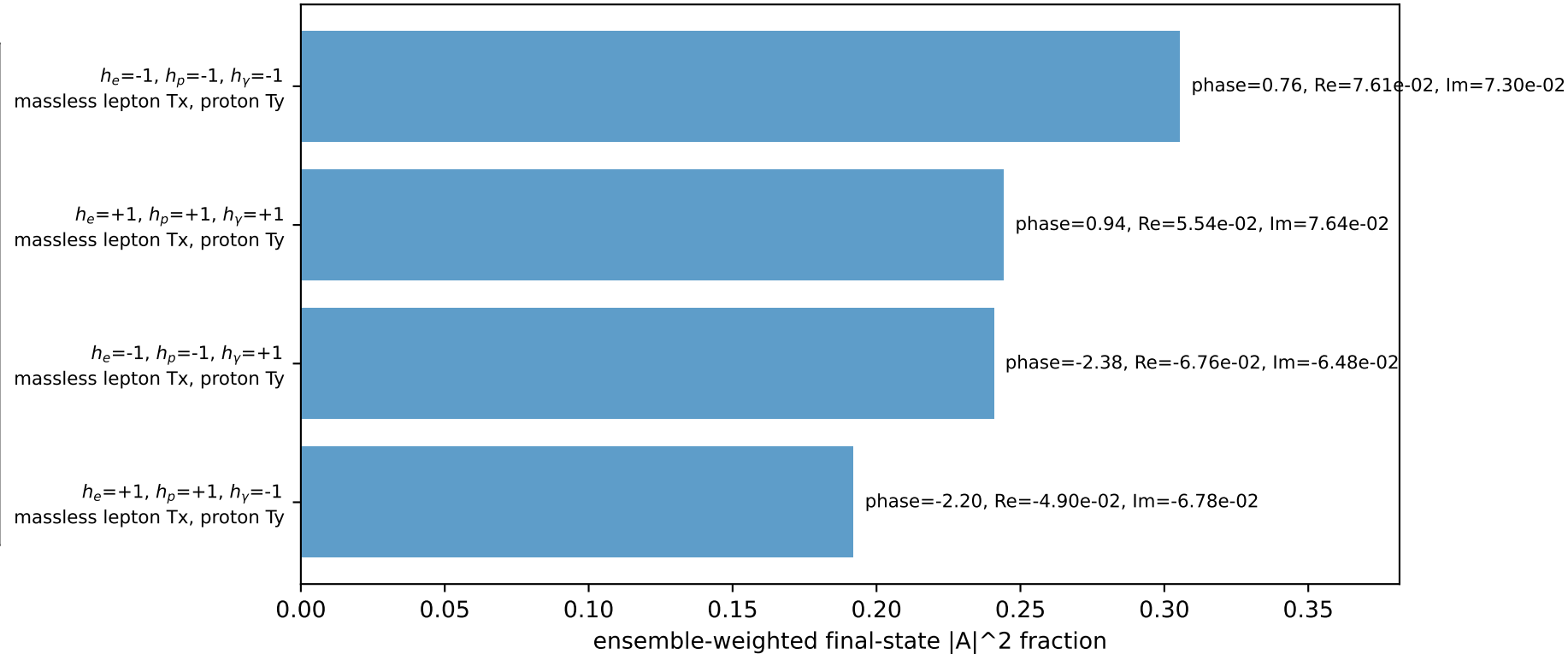
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=2.2216$
 $\theta_{\text{in}}=1.57079$, $\phi_{\ell}=0.883573$, $\phi_P=4.02517$
 $E_{\gamma}=0.25$, $\phi_{\gamma}=4.51604$
 $Q^2=15.4546$, $x_B=0.870691$, $t=-17.5236$, $W^2=3.17506$, $y=0.860138$
 $\theta(\ell', \gamma)=12.6636$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2244$ $p_x= 1.4112$ $p_y= 1.7195$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= -1.4112$ $p_y= -1.7195$ $p_z= 0.0000$
 ℓ' $E= 1.9770$ $p_x= 0.0488$ $p_y= -1.9764$ $p_z= 0.0000$
 P' $E= 2.4115$ $p_x= 0.0000$ $p_y= 2.2216$ $p_z= 0.0000$
 q_{γ} $E= 0.2500$ $p_x= -0.0488$ $p_y= -0.2452$ $p_z= 0.0000$

Transverse plane

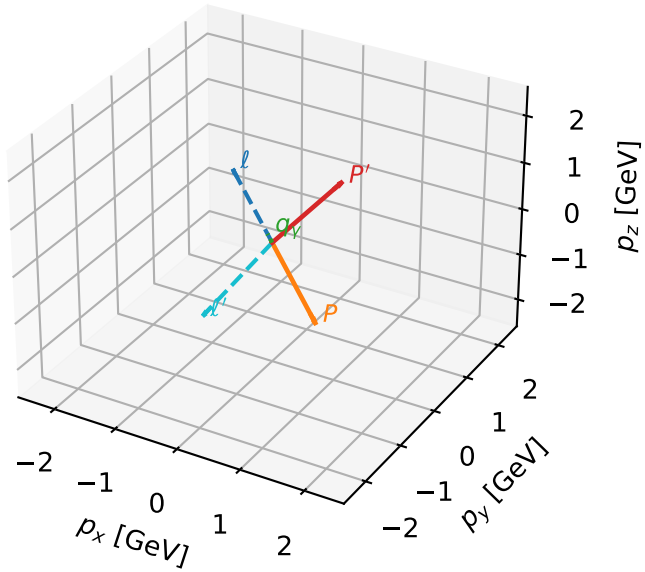


Leading initial-ensemble/final-state components (N=4, retained=98.2%)



Tx massless lepton + Ty proton characteristic max P_{GHZ} + configuration

3D momenta



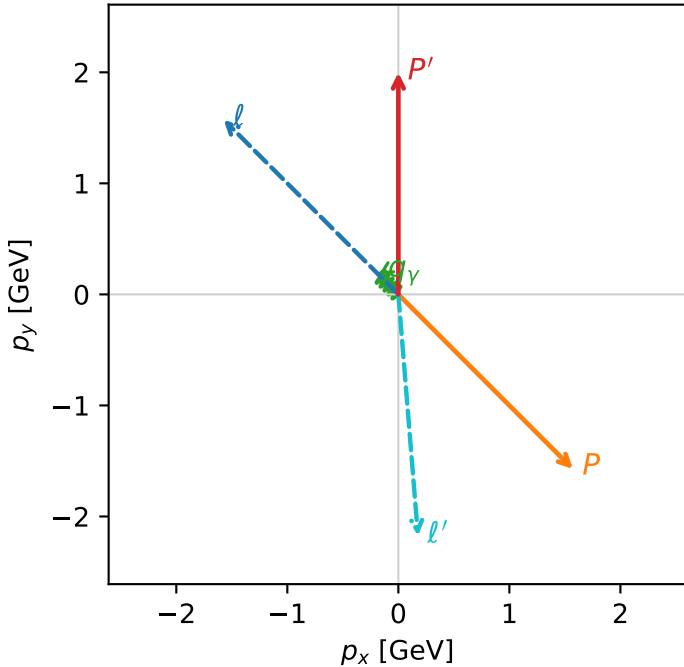
ghz_purity_low_egamma_txtty_region_3 $P_{\text{GHZ}}=0.542591$
 region: low_Egamma_TxTy_region_3
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_low_Egamma
 incoming state: $|T_{X,\ell_0}\rangle \otimes |\bar{T}_{Y,p}\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=1.99765$
 $\theta_{\text{in}}=1.57079$, $\phi_{\ell}=2.35619$, $\phi_P=5.49779$
 $E_{\gamma}=0.25$, $\phi_{\gamma}=2.35619$
 $Q^2=17.102$, $x_B=0.977303$, $t=-15.18$, $W^2=1.27702$, $y=0.847994$
 $\theta(\ell', \gamma)=139.648$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ	$E=$	2.2244	$p_x=$	-1.5729	$p_y=$	1.5729	$p_z=$	-0.0000
P	$E=$	2.4141	$p_x=$	1.5729	$p_y=$	-1.5729	$p_z=$	0.0000
ℓ'	$E=$	2.1816	$p_x=$	0.1768	$p_y=$	-2.1744	$p_z=$	0.0000
P'	$E=$	2.2069	$p_x=$	0.0000	$p_y=$	1.9977	$p_z=$	0.0000
q_{γ}	$E=$	0.2500	$p_x=$	-0.1768	$p_y=$	0.1768	$p_z=$	0.0000

Transverse plane



Leading initial-ensemble/final-state components (N=4, retained=97.7%)

